Joey Chung

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EDUCATION

University of Michigan, Ann Arbor *Aug. 2025 – Present*
B.S. in Chemistry | GPA: 3.99 / 4.00

University of Wisconsin, Madison *Sept. 2022 – May 2024*
Coursework toward B.S. in Food Science | GPA: 3.96 / 4.00
Honors: Dean's List (4 semesters); Norman F. Olson Family Scholarship (2024); Bill Reese Dairy, Deli & Bakery Study Grant (2023); Wisconsin Alumni Association Korean Chapter Scholarship (2022)

PROFESSIONAL EXPERIENCE

DAEYANG Co. Ltd | Research Assistant *Daegu, South Korea | Sept. 2024 – May 2025*
Corporate research institute developing thermoelectric materials and modules for energy applications (re-dy.com/en)

- Ran medium phosphorus (6 to 8% P) electroless nickel plating on bismuth telluride pellets, including alcohol-based cleaning and pH monitoring and adjustment. After identifying a plating stability issue, proposed and designed a custom titanium holding jig with full technical drawings and specifications; implementation stalled on production scheduling, fabricator sourcing, and MOQ constraints.
- Inspected diced thermoelectric legs against design tolerances, classifying defects by type (machining error, burring, asymmetry). On a module build commissioned by the Korea Electrotechnology Research Institute (KERI), separately handled incoming inspection of the supplied legs and tray loading, pulling nonconforming units before SMT.
- Supported senior engineers on SMT assembly of 8×8 thermoelectric modules (128 legs each): solder paste prep, mounting frame setup, substrate loading, and soldering positive and negative lead wires to module connectors after reflow. Weekly output ran 20 to 30 modules, scaling past 50 during government R&D cycles.
- Ran final electrical inspection on finished modules, measuring resistance and voltage against spec, documenting results in Excel for traceability, and flagging nonconforming units for retest.
- Collected thermoelectric generator performance data across five cooling configurations (forced and natural air, forced and natural water, antifreeze based) to support comparative analysis.

Military Service | Auxiliary Police Officer (Sergeant) *Pohang, South Korea | Nov. 2019 – May 2021*
Mandatory service under the Ministry of National Defense and the Korean National Police Agency

- Led a squad of 12 officers for 7 of the 18-month term through high pressure field deployments, including protest response and missing person searches across mountainous and coastal terrain; located 3 missing individuals and escorted them to safety, earning an official commendation.
- Managed police lines during civil disturbances and took part in joint readiness drills with the ROK Marine Corps, including chemical threat response protocols. Staffed station entry control, guiding civilians to the correct department, and provided driving support to transport personnel and equipment.

RESEARCH & TEACHING

Maldonado Group | Undergraduate Research Assistant *UM-Ann Arbor | April 2026 – Present*
Advisor: Prof. Stephen Maldonado

- Take part in the group's Atomic Layer Deposition (ALD) work, operating the system to grow thin films to target thickness. Characterize the resulting layers by ellipsometry and Atomic Force Microscopy (AFM), measuring film thickness and surface step height, and document run conditions and results for group review.

Organic Chemistry Peer Mentor | Academic Support *UW-Madison | Fall 2023 – Spring 2024*

- Resolved 100+ academic inquiries on Piazza and led interactive sessions on organic chemistry concepts and multistep reaction mechanisms.

LEADERSHIP & GLOBAL ENGAGEMENT

Michigan Rifle Team | Member *UM-Ann Arbor | Sept. 2025 – Present*

- Competed in three-position and air rifle events (Camp Perry Open, WIRC Championship); created social media photo content and organized team building events.

Food Science Club | Member *UW-Madison | Sept. 2022 – Dec. 2023*

- Coordinated HR led recruitment sessions with industry representatives and organized collaborative cooking activities to boost member engagement.

International Food and Cultural Engagement | Cultural Outreach *8+ Countries | Jan. 2021 – Present*

- Cooked Korean fusion dishes (e.g., bulgogi tacos, gochujang-based sauces) for friends and communities through informal cooking sessions and festival style events across countries including Spain, Poland, and Mexico.

SKILLS

- **Process & Quality Engineering:** SMT workflow, parameter control, solder paste and surface prep, fixture design, dimensional inspection, defect classification, comparative evaluation, data logging and traceability, Excel reporting, power calculations.
- **Materials & Surface Characterization:** Atomic Layer Deposition (ALD), ellipsometry, Atomic Force Microscopy (AFM).
- **Leadership & Safety:** Safety first execution, checklist readiness, high pressure coordination, clear technical communication.